Practical 1a: MVC Application for Basic Arithmetic Operations

AIM To create an ASP.NET Core MVC application that performs basic arithmetic operations (Addition, Subtraction, Multiplication, Division) using C#.

Step 1:

1.Create Project Open Visual Studio → File → New → Project
2. Select 'ASP.NET Core Web App (Model-View-Controller)'
3. Name the project: ArithmeticMVC → Click Create
4. Select Framework: .NET 6.0 (or .NET 8.0) → Click Create

Step 2:

Create the Model Right-click Models folder → Add → Class → Name: ArithmeticModel.cs

Program Code

Models/ArithmeticModel.cs

```
using System.ComponentModel.DataAnnotations;
 namespace ArithmeticMVC.Models {
   public class ArithmeticModel
      {
        [Required]
        [Display(Name = "First Number")] public double Number1 { get; set; }
        [Required]
        [Display(Name = "Second Number")] public double Number2 { get; set; }
        public string Operation { get; set; } = "";
        public double Result { get; set; }
        public string ErrorMessage { get; set; } = "";
      }
  }
```

Controllers/HomeController.cs

```
using ArithmeticMVC.Models;
using Microsoft.AspNetCore.Mvc;

namespace ArithmeticMVC.Controllers
{
   public class HomeController : Controller
```

```
    {
        public IActionResult Index()
        {
            return View(new ArithmeticModel());
        }

        [HttpPost]
        public IActionResult Index(ArithmeticModel model)
        {
            if (ModelState.IsValid)
            {
                switch (model.Operation)
                {
                    case "Add":
                        model.Result = model.Number1 + model.Number2;
                        break;
                    case "Subtract":
                        model.Result = model.Number1 - model.Number2;
                        break;
                    case "Multiply":
                        model.Result = model.Number1 * model.Number2;
                        break;
                    case "Divide":
                        if (model.Number2 == 0)
                        {
                            model.ErrorMessage = "Error: Division by zero!";
                        }
                        else
                        {
                            model.Result = model.Number1 / model.Number2;
                        }
                        break;
                }
            }

            return View(model);
        }
    }
}
```

Views/Home/Index.cshtml

```
@model ArithmeticMVC.Models.ArithmeticModel
@{
   ViewData["Title"] = "Arithmetic Operations";
}

<h2 class="text-center mt-3">Basic Arithmetic Operations</h2>
<hr />

<div class="container">
   <form asp-action="Index" method="post">
      <div class="row justify-content-center">
         <div class="col-md-6">
            <div class="mb-3">
               <label asp-for="Number1" class="form-label"></label>
               <input asp-for="Number1" class="form-control" />
               <span asp-validation-for="Number1" class="text-danger"></span>
            </div>
            <div class="mb-3">
               <label asp-for="Number2" class="form-label"></label>
               <input asp-for="Number2" class="form-control" />
               <span asp-validation-for="Number2" class="text-danger"></span>
            </div>
            <div class="mb-3">
               <button type="submit" name="Operation" value="Add" class="btn btn-
primary">Add (+)</button>
               <button type="submit" name="Operation" value="Subtract" class="btn btn-
warning">Subtract (-)</button>
               <button type="submit" name="Operation" value="Multiply" class="btn btn-
success">Multiply (*)</button>
               <button type="submit" name="Operation" value="Divide" class="btn btn-
danger">Divide (/)</button>
            </div>

            @if (Model.ErrorMessage != "")
            {
               <div class="alert alert-danger">@Model.ErrorMessage</div>
            }
            else if (Model.Operation != "")
```

```
            {
                <div class="alert alert-success">
                    <strong>Result:</strong> @Model.Number1 @Model.Operation
@Model.Number2 = @Model.Result
                </div>
            }
        </div>
    </div>
  </form>
</div>
```

Output:

